

Aditya Sugandhi

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SUMMARY

Software Engineer with 5+ years building 0→1 AI products and generative AI solutions from ideation through production. Proven track record prototyping and pathfinding new opportunities with LLMs, RAG pipelines, and AI agents, delivering measurable business value through deep product and stakeholder collaboration. Production experience implementing full lifecycle AI development—from model selection and prompt engineering to productionizing, monitoring, and maintaining AI systems at scale.

WORK EXPERIENCE

Software Engineer — Enterprise Applications & AI
Florida State University

Nov 2024 – Present
Tallahassee, FL

- **Led 0→1 generative AI product development** from ideation to production: worked embedded with stakeholders to identify highest-impact use cases, prototyped **LLM-based solutions** with multiple model architectures (GPT-4, Llama 3.3), and productionized **AI agents with RAG pipelines** that reduced manual analysis time by **40%** across **172+ networks**.
- **Implemented full lifecycle AI development**: evaluated and selected models based on latency/accuracy tradeoffs, built **prompt engineering frameworks** with version control, deployed **production monitoring** (accuracy tracking, hallucination detection), and maintained systems with **99.2% reliability** serving daily operational workflows.
- Developed **deep product and business knowledge** by embedding with facility managers and operations teams; translated abstract AI capabilities into **tangible business value**, connecting **semantic search and document analysis** to **85% reduction in manual processes** and **\$200K+ annual operational savings**.
- Built **production-grade AI product** with **React/TypeScript frontend** and **Python backend (37 REST API endpoints)**; shipped **real-time Web-Socket streaming** for AI responses, designed **user-centered interfaces** for non-technical operators, and established **CI/CD pipelines** with automated testing for AI components.

Software Engineer — Data Platform & ML Systems
Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024
Tallahassee, FL

- **Prototyped and pathfound generative AI opportunities** on **1TB+ unstructured datasets**: experimented with **vector embedding models** (OpenAI, Sentence-BERT), built **semantic search prototypes**, and validated **RAG architecture patterns** that became foundation for production AI systems processing **1,000+ documents** with **95% accuracy**.
- Implemented **early-phase AI development lifecycle**: evaluated **5+ embedding models** and **3 vector databases** (Pinecone, Weaviate, ChromaDB) through rapid prototyping, selected optimal architecture based on **retrieval precision and latency (sub-200ms)**, and productionized with monitoring and feedback loops.
- Connected **AI modeling tasks to business outcomes**: collaborated cross-functionally with researchers and domain experts to translate **complex analytical problems** into **operationalized AI workflows**, delivering **50% pipeline runtime reduction** through iterative optimization and stakeholder feedback.
- Built **production ML/LLM infrastructure** on **Kubernetes** with **distributed compute (Spark/Python)**; designed **data pipelines** for AI model training and inference, achieving **+40% throughput** through custom partitioning and query optimization on large-scale datasets.

Software Engineer — Enterprise Data Integration
Aspire Systems

Oct 2020 – Jul 2022
Chennai, India

- Delivered **end-to-end enterprise software solutions** using **Java/Spring Boot** and **React**; built **event-driven microservices** with **Kafka** processing **500K+ events/day**, demonstrating **full-stack product development** skills from backend architecture to user-facing interfaces with **99.5% availability**.
- Developed **product and business acumen** by working closely with stakeholders across departments; identified high-value problems, designed **data integration workflows** connecting enterprise systems, and delivered solutions that enabled **automated decision-making** with measurable operational impact.
- Built **production monitoring and observability infrastructure** with **Prometheus, Grafana, and distributed tracing**; took full ownership of **system maintenance and incident response**, reducing **MTTR by 60%** through proactive monitoring and rapid troubleshooting.
- Collaborated with **cross-functional teams** and mentored **5 engineers** on best practices; shipped iteratively with rapid feedback cycles, achieving **90% test coverage** and maintaining high code quality standards across distributed systems.

PROJECTS

- **AI-Powered Resume Optimization Platform** [GitHub](#) — 0→1 generative AI product: prototyped and built **full-stack AI application** from scratch in **React/TypeScript** and **Next.js**. Implemented **full AI lifecycle**—evaluated **LLMs (Llama 3.3, GPT-4)**, built **RAG pipelines** with **vector embeddings**, deployed **AI agents** for document analysis, and productionized with monitoring. Delivered end-to-end: **semantic search**, prompt engineering, and production UI. Tech: **TypeScript, React, Python, Groq API, Vector DBs**.
- **Enterprise Slack Bot** — **AI Workflow Automation** [GitHub](#) — **Productionized LLM-based solution** as enterprise Slack integration; implemented **full lifecycle development** from model selection to deployment and maintenance. Built **AI agent** with tool integrations, achieved **24/7 availability** and **sub-40s response times**, deployed on **AWS EC2** with monitoring. Tech: **Python, Node.js, OAuth 2.0, WebSockets, PM2**.
- **Distributed Data Pipeline (Spark/Kafka/Kubernetes)** [GitHub](#) — Architected **real-time data ingestion platform** with **Spark** and **Kafka** on **Kubernetes**; built orchestration layer for multi-source enterprise data integration processing **10K+ events/sec**. Demonstrated infrastructure skills for **scaling AI/ML workloads**. Tech: **Java, Scala, Spark, Kafka, Kubernetes, Docker**.

TECHNICAL SKILLS

- **AI/ML & Generative AI:** LLMs (GPT-4, Llama, Claude), RAG Pipelines, AI Agents, Vector Databases (Pinecone, Weaviate, ChromaDB), Embedding Models, Prompt Engineering, Model Evaluation, MLflow, Full Lifecycle AI Development
- **Languages:** Python, TypeScript, JavaScript, Java, Scala, SQL, Go, Bash, C++
- **Frontend & Product:** React, Next.js, TypeScript, HTML/CSS, Responsive UI, Component Libraries, Production Frontend Applications, User-Centered Design
- **Data & Platforms:** Apache Spark (PySpark/Scala), Kafka, PostgreSQL, Redis, MongoDB, Snowflake (familiar), Data Pipelines, Real-time Streaming
- **Infrastructure & DevOps:** Kubernetes (EKS/AKS), Docker, AWS (EKS, EMR, Lambda, S3, EC2), CI/CD (GitHub Actions, Jenkins), Grafana, Prometheus, Production Monitoring
- **Practices:** 0→1 Product Development, Rapid Prototyping, Stakeholder Collaboration, Agile/Scrum, End-to-End Ownership, TDD, Code Review, Technical Documentation

EDUCATION

Florida State University
M.S. in Computer Science

Tallahassee, FL

- **Relevant Coursework:** Distributed Systems, Advanced Database Systems, Machine Learning, Data Communications, Algorithms

SRM Institute of Science & Technology
B.Tech. in Computer Science

Chennai, India

PUBLICATIONS & AWARDS

- **2nd Place, SRM University Hackathon (2019)** — Built end-to-end data-driven application in 24 hours; designed ML-based review classification system from problem discovery to working demo, judged on business impact and technical execution.
- **Review Classification & False Feedback Detection** — Published in IJAST; built production ML pipeline for large-scale unstructured text analysis.
- **AWS Solutions Architect Course (Udemy, 2024)** | Active participant in AI/ML hackathons and rapid prototyping challenges.